

Principles of Microeconomics

Day 1 — Costs of Production

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Where Day 1 sits in the week

The plan

- **Day 1 (today) — Costs.** What it costs the firm to make output.
- Day 2 — Perfect competition: the firm as a price taker.
- Day 3 — Monopoly: a single price maker.
- Day 4 — Duopoly: two firms competing.

Costs come *first* because every decision in Days 2–4 — produce, shut down, enter, exit, set a price — compares revenue against **cost**. Get the cost side right today and the rest of the week is just applying it.

What you will be able to do by the end

Learning goals

- ① Say *why* a firm must know its cost function.
- ② Define FC, VC, TC, AFC, AVC, ATC and MC.
- ③ Derive each of them from a total-cost formula.
- ④ Explain why MC cuts AVC and ATC at their lowest points.
- ⑤ Explain why fixed cost vanishes in the long run.

Nothing here is memorisation — every curve today comes from one cost formula.

The firm's basic question

A firm wakes up and asks two things:

- ① *Should I produce at all?*
- ② *If so, how much?*

Both answers need the same ingredient: how cost changes with output.

We cannot answer either until we can write cost as a function of quantity, $TC(Q)$.

Profit splits into two halves

The identity behind everything

$$\pi(Q) = \underbrace{TR(Q)}_{\text{revenue side}} - \underbrace{TC(Q)}_{\text{cost side}}$$

- Revenue depends on the market (Days 2–4).
- Cost depends on the firm's technology — **that is today**.
- You cannot maximise a difference if you only know one of its two terms.

Two lenses on cost: marginal and average

Marginal cost

The cost of *one more* unit.

- Answers: “is the next unit worth making?”

Average cost

Cost *per unit* produced.

- Answers: “is the whole operation worth running?”

Keep these two apart. Confusing them is the single most common mistake in this topic.

Two time horizons — the idea

The words “short” and “long” are about **commitment**, not the calendar.

- **Short run:** at least one input is *fixed* — the firm is stuck with it.
- **Long run:** *every* input can be changed — nothing is locked in.

A restaurant can hire another cook tonight (variable) but is stuck with its building until the lease ends (fixed). The long run is the horizon over which even the lease can be undone.

Short run vs long run — side by side

Short run

- Plant size fixed (capital).
- Cannot escape the fixed cost.
- Adjusts output via labour, materials.

Long run

- Plant size chosen freely.
- **No fixed cost** — can exit entirely.
- Picks the best plant for each output.

We spend most of today in the *short run*, then look at the long run at the end.

Fixed cost

Fixed cost (FC)

Cost that does *not* change with output — paid even when $Q = 0$.

- Rent on the building, insurance, a machine already bought.
- Produce 0 units or 1000 units: the same FC is owed.
- In symbols, FC is a constant.

Variable cost

Variable cost (VC)

Cost that *rises with output* — and is zero when nothing is produced.

- Raw materials, hourly wages, electricity for the machines.
- $VC(0) = 0$: make nothing, spend nothing on variables.
- Written as a function of quantity, $VC(Q)$.

Total cost

Total cost (TC)

$$TC(Q) = FC + VC(Q)$$

- At $Q = 0$: $VC = 0$, so $TC = FC$. The firm still owes its fixed cost.
- As Q rises, TC and VC climb together — they differ only by the constant FC .

So TC is just the VC curve lifted up by FC .

From totals to per-unit measures

We now divide the totals by output to get *per-unit* cost.

Three averages

$$AFC = \frac{FC}{Q} \quad \text{(average fixed cost)}$$

$$AVC = \frac{VC(Q)}{Q} \quad \text{(average variable cost)}$$

$$ATC = \frac{TC(Q)}{Q} = AFC + AVC \quad \text{(average total cost)}$$

The last equality just splits total average cost into its two pieces.

Average fixed cost falls — always

$$AFC = \frac{FC}{Q}$$

A fixed number divided by a rising output.

- As Q grows, AFC keeps shrinking — “spreading the overhead”.
- It never reaches zero, but gets arbitrarily small.
- This falling AFC is what pulls ATC down at low output.

Marginal cost

Marginal cost (MC)

The extra cost of producing *one more* unit:

$$MC = \frac{dTC}{dQ} = \frac{dVC}{dQ}.$$

- FC drops out — it does not change, so it adds nothing at the margin.
- MC depends only on the *variable* cost.
- This is the number that decides whether the next unit is worth making.

A concrete cost function

Take one firm and stick with it for the next few slides:

$$VC(Q) = \frac{1}{3}Q^3 - 4Q^2 + 20Q, \quad FC = 40.$$

So $TC(Q) = \frac{1}{3}Q^3 - 4Q^2 + 20Q + 40$.

We will derive MC , AVC , AFC and ATC one at a time — no shortcuts.

Step 1 — marginal cost

Differentiate TC with respect to Q :

$$MC = \frac{dTC}{dQ} = Q^2 - 8Q + 20.$$

- Note the $FC = 40$ disappeared: constants differentiate to zero.
- Quick check: $MC(6) = 36 - 48 + 20 = 8$.

Step 2 — average variable cost

Divide VC by Q :

$$AVC = \frac{VC}{Q} = \frac{1}{3}Q^2 - 4Q + 20.$$

- Its lowest point: set $\frac{d AVC}{d Q} = \frac{2}{3}Q - 4 = 0 \Rightarrow Q = 6$.
- Value there: $AVC(6) = 12 - 24 + 20 = 8$.
- Notice $MC(6) = 8$ too — we return to this.

Step 3 — average fixed and average total cost

$$AFC = \frac{40}{Q}, \quad ATC = AVC + AFC = \frac{1}{3}Q^2 - 4Q + 20 + \frac{40}{Q}.$$

- ATC is just the two curves added vertically.
- At small Q , the $40/Q$ term dominates, so ATC starts high.
- As Q grows, $40/Q$ fades and ATC approaches AVC .

The example in numbers

Evaluating each formula at a few output levels:

Q	MC	AVC	AFC	ATC
2	8.00	13.33	20.00	33.33
4	4.00	9.33	10.00	19.33
6	8.00	8.00	6.67	14.67
8	20.00	9.33	5.00	14.33

Watch row $Q = 6$: $MC = AVC = 8$. And ATC is still falling past $Q = 6$ (it is lower at $Q = 8$).

Why MC passes through the bottom of AVC

The averaging rule

An average *falls* when the next value is below it, and *rises* when the next value is above it.

- If $MC < AVC$: the new unit costs less than the running average, so AVC falls.
- If $MC > AVC$: the new unit costs more, so AVC rises.
- At the turning point — the bottom of AVC — we must have $MC = AVC$.

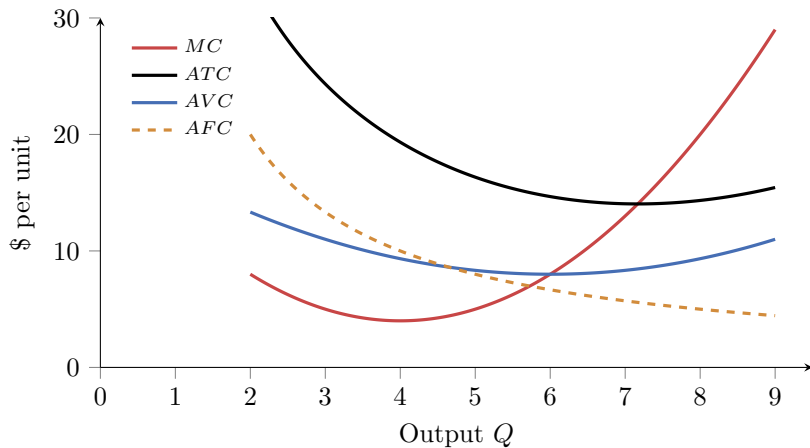
The same logic for ATC

Two facts to remember

- ① MC cuts *both* AVC and ATC at their minimum points.
- ② The minimum of ATC lies to the *right* of the minimum of AVC .

Why (2)? Just past the bottom of AVC , AVC is rising only gently, and AFC is *still* falling. The falling AFC keeps ATC declining a little longer — so ATC bottoms out later.

The cost curves, drawn



What to notice in the picture

- AFC (dashed) slides down toward zero — overhead being spread.
- AVC and ATC are U-shaped; the gap between them *is* AFC .
- MC slices through the bottom of AVC , then through the bottom of ATC .
- The ATC trough sits to the right of the AVC trough — exactly as predicted.

These curves reappear in every remaining session this week.

In-slide exercise (no solution given)

Your turn

A firm has $TC(Q) = 2Q^2 + 8Q + 50$.

- 1 Identify FC and $VC(Q)$.
- 2 Write AFC , AVC , ATC and MC .
- 3 Does MC lie above or below AVC here? What does that tell you about AVC ?

Work it in pairs; we compare answers before the break.

Long run: where did the fixed cost go?

In the long run *every* input is variable — including the plant.

Consequence

There is no input the firm is stuck with, so there is **no fixed cost**.

- Every long-run cost is a variable cost.
- The firm can also choose *not* to operate at all (exit).

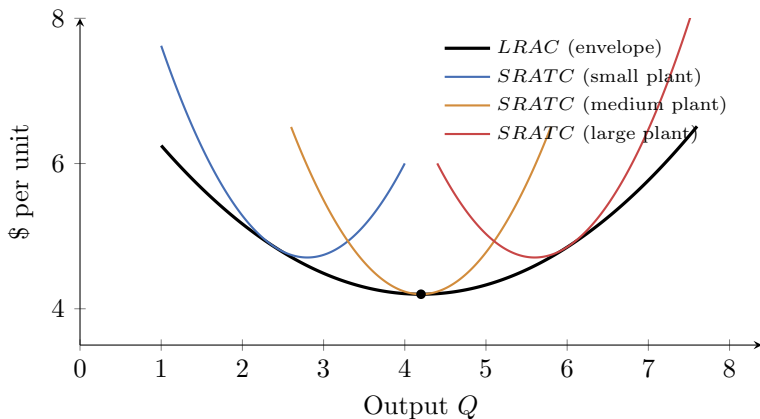
Choosing the right plant

The long-run idea

For each output level, the firm picks the plant size that makes that output as cheap as possible.

- Each possible plant has its own short-run ATC curve.
- The firm gets to pick the best one for wherever it wants to produce.
- So the long-run average cost is built from the *lowest* points available across all plants.

Long-run average cost as an envelope



Reading the envelope

- Each coloured curve is one plant's short-run ATC .
- The black $LRAC$ hugs them from below — it touches each where that plant is best.
- $LRAC$ is never above any short-run curve: in the long run you can always re-pick the plant.
- The dot marks the cheapest point of all — the *minimum efficient scale*.

Economies and diseconomies of scale

Along the $LRAC$ curve:

- **Falling** $LRAC$ = *economies of scale* — bigger is cheaper per unit.
- **Flat** $LRAC$ = constant returns — size does not change unit cost.
- **Rising** $LRAC$ = *diseconomies of scale* — too big to manage cheaply.

The shape of $LRAC$ shapes how many firms a market can support — a theme in Days 2–3.

Day 1 — checkpoint

You should now be able to

- split TC into $FC + VC$ and build all six per-unit curves;
- differentiate TC to get MC and know why FC drops out;
- explain why MC cuts AVC and ATC at their minima;
- explain why ATC 's minimum is to the right of AVC 's;
- say why fixed cost disappears in the long run.

Next session we add the *revenue* side and let the firm choose output.

In-class exercise

Exercise 1 — Costs of Production

Work through **Exercise 1** on the accompanying sheet: a full cost function with a cubic TC , where you find FC , all the averages and MC , locate the minimum of AVC , and confirm $MC = AVC$ there.

Different numbers from today's example — fresh practice, same method.